NOTICE OF INTENT

This document provides information concerning the major tasks and typical responsibilities associated with installation of GE supplied equipment. It describes the overall work scope, but does not detail specific methods. Its primary purpose is to provide an overview of a typical installation process. Design changes subsequent to the issue date of this document may not be reflected in this guide.

This document does not purport to cover all aspects of a particular installation. Unit-specific details are provided in the document and drawing packages supplied with each requisition, and all sketches and descriptions are to be considered "Typical." This document is property of General Electric and is considered Company Proprietary Information.

HEALTH AND SAFETY REQUIREMENTS

The GI&FS Environmental Health & Safety Manual (EHSM) outlines the requirements that must be followed on the job site during an installation or maintenance outage.

Safety, health, and protection of the environment must be an integral part of all business operations, design, construction and start-up, and includes EHS planning prior to and during work to anticipate and minimize hazards and incidents. Every reasonable effort must be made to create and maintain a safe and healthy work environment at each customer site.

Where working conditions are not covered by these safety requirements and procedures, or when a requirement or a procedure is not completely understood, obtain specific instructions or clarification from manager, supervisor or EHS before proceeding with the work.

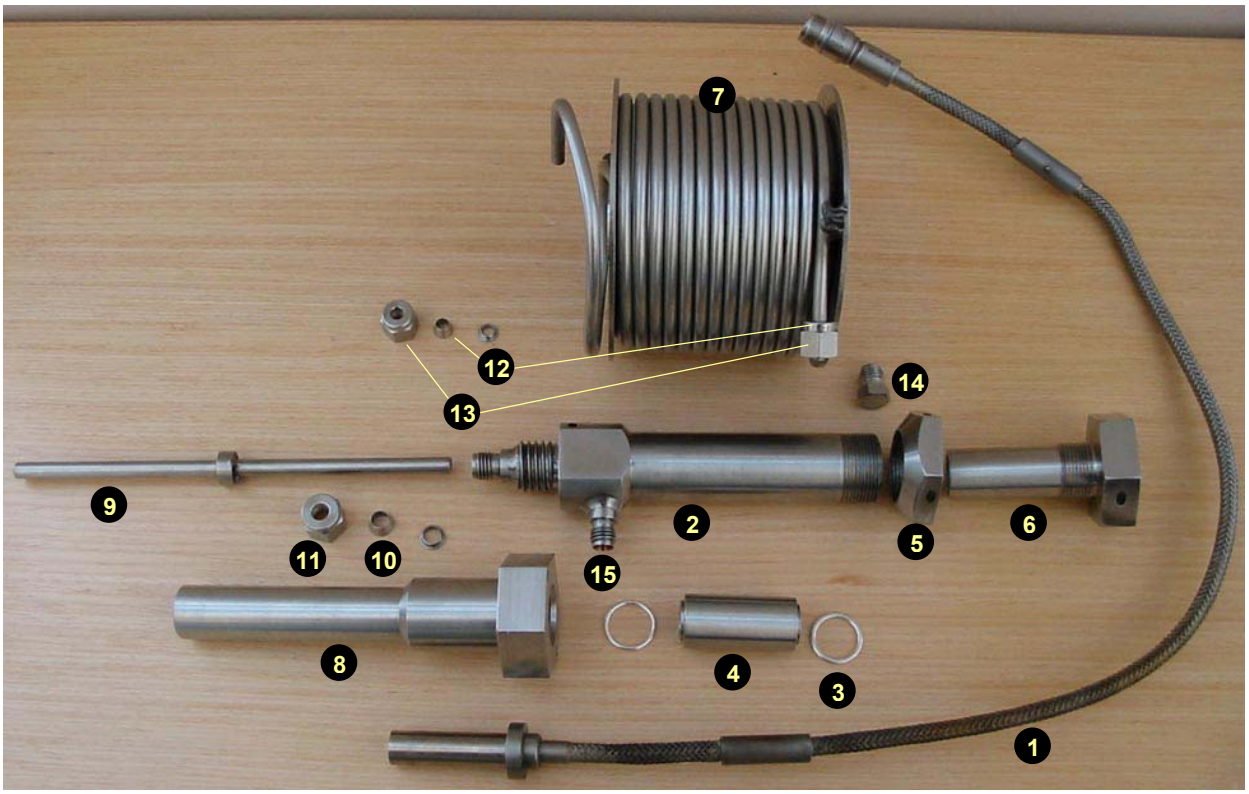
REFERENCE DOCUMENTS

1.) MLI 0726 Dynamics Probe Setting Tool

2.) MLI 0726 Procedure for using Dynamics Probe Depth Gauge

To identify the specific reference drawings for your unit, connect to the Unit Record Sheet Web page: http://nyschi52psge.sch.ge.com/eng/Projects/copicsURS/choice.cfm?lnkchoice=3

PRESSURE TRANSDUCER INSTALLATION



Pressure Transducer Parts List

Index	Description	Qty.
1	High Temperature probe	1
2	Sensor Housing	1
3	Inconel O-Ring	2
4	Acoustic Sleeve	1
5	Housing Chamfered Nut	1
6	Housing Flange	1
7	Dampening Coil Wound on metal Spool	1
8	Combustor tip Support	1
9	Measurement tip	1
10	1/4 Two Piece Ferrule	1
11	1/4 Swagelok Nut	1
12	3/16 Two Piece Ferrule	2
13	3/16 Swagelok Nut	2
14	Tube plug	1
15	Port	1

GEN II PROBE HOLDER ASSEMBLY INSTRUCTIONS

Please Read through and understand these instructions before assembling the Gen II Probe Holder. Contact the Remote Services team with any questions.

1. If not already done install plug at the straight end of 3/16 tubing. Tighten Swagelok Fitting using the correct Swagelok tightening procedures, (refer to Reference Document 2 and Figure 1).
2. Slide the Dampening coil onto the sensor housing. (Figure 2)
4. Slide the 3/16 Nut and Ferrell over the tubing on bent the Dampening Coil. (Figure 3)
5. Insert the 3/16 tubing from the dampening coil into the port of Sensor housing. (Figure 4)



Figure 1

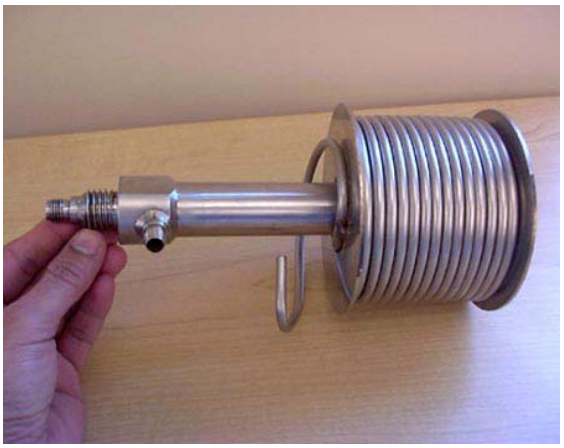


Figure 2



Figure 3



Figure 4

6. Push hard until tubing bottoms out. Then while maintaining pressure, tighten 3/16 Swagelok Fitting. (See Figure 5) Use the correct Swagelok tightening procedures, found on Reference Document 2.

7. Hold the damping coil in place using the Chamfered Nut. Use high temperature anti-seeze compound on treads. (See Figure 6.) The chamfered nut should be tightened to 70 ft-lbs ensure the coil will be held securely in place.

8. Be sure that nut is correctly treaded perpendicular to the sensor housing it is easy to cross tread. (See Figure 7.)

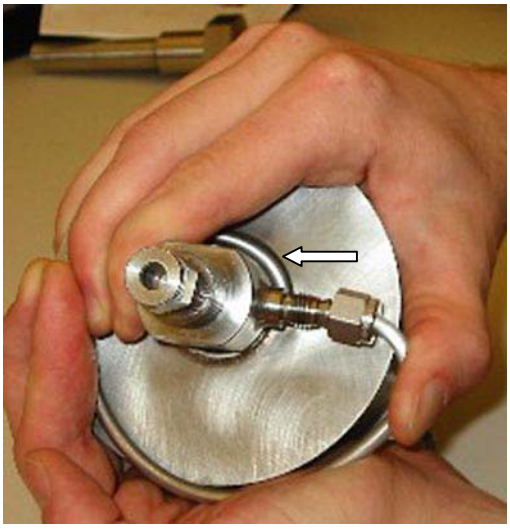


Figure 5



Figure 6



Correct



Incorrect

Figure 7

9. Place the 1st O-Ring in the upper groove of the sleeve. Place the 2nd O-Ring in the lower groove of the sleeve. (Figure 8)

Caution **Ensure Both O-Rings Are Used**

Caution **Use NEW O-Rings each time probe holder is assembled. O-rings cannot be reused.**

10. Slide the Sleeve inside the Sensor Housing. Make sure the Inconel O-Rings correctly seat in the sleeve groove. (Figure 9)

11. Place the Combustion Dynamics Sensor in the Sensor Housing. (Figure 10)

12. Slide the Housing flange over the Probe and hold in place with the Sensor Housing. (Figure 11)



Figure 8



Figure 9

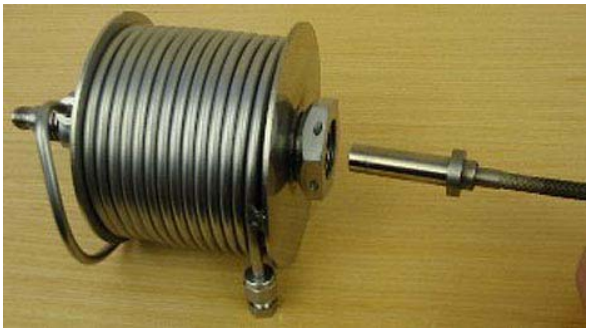


Figure 10

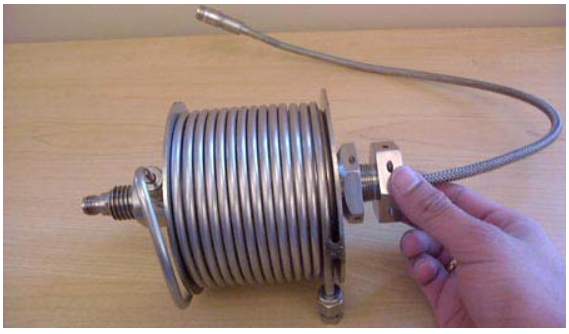


Figure 11

The housing flange is reverse treaded, be sure to use High Temp anti-seize compound. Tighten the Housing flange to (25 FT-LBS). This step is critical to ensure correct O-ring crush.

If installed correctly there should be a gap between the housing flange and the chamfered nut. (See Figure 12)

13. Slide ¼ Swagelok nut and two-piece Ferrell assembly over the Measurement tip. (Figure 13)

Caution **Ensure the Measurement tip is the correct length for the specific turbine application. See Drawing 241C2842 for length details.**

Make sure to insert the Measurement Tip fully into the other end of Sensor Housing.

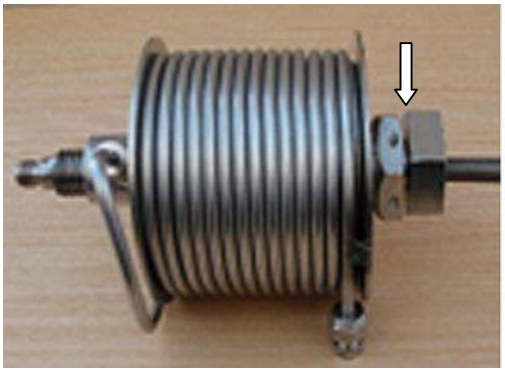


Figure 12



Figure 13

14. Tighten 1/4 Swagelok nut using the correct Swagelok tightening procedures this can be found on Reference Document 2. (Figure 14)

15. Apply Loctite Heavy Duty Anti-Sieze evenly to the 3/4 –10 treads of the sensor housing.

Slide the Combustor tip support over the Measurement Tip and fasten to the Sensor Housing. Tighten the 1-1/2 inch nut to 115 ft-lbs. Shown in Figure 15 without damping coil for reference.

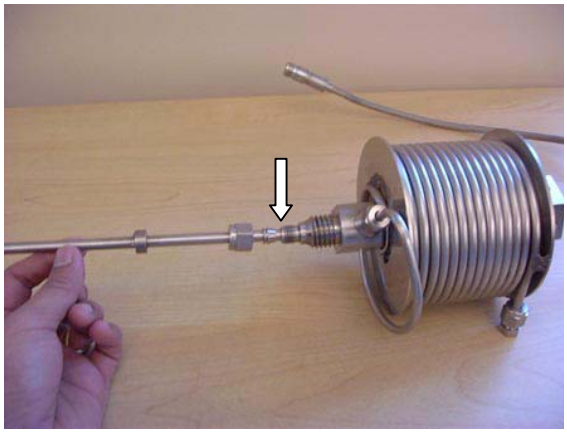


Figure 14

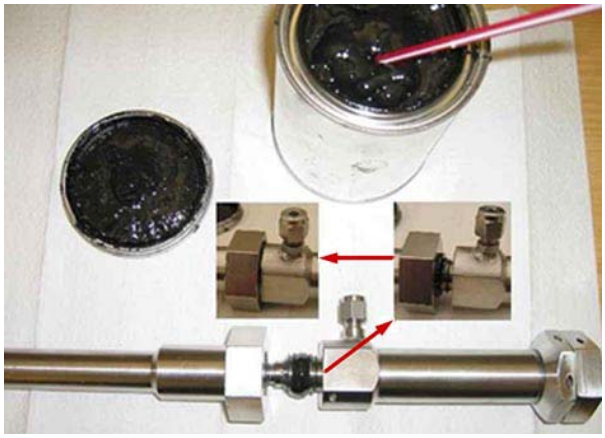


Figure 15

16. Using Wire ties secure the assembly in the locations show in Figure 16.

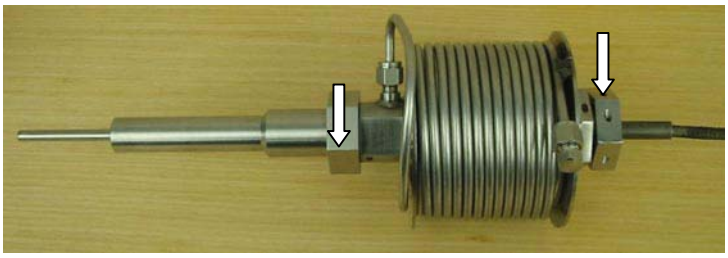


Figure 16

Assembly Suggestion:

When setting the ¾" Swagelok Ferrule on the Gen II Probe Holder it may be easier to first install the measurement tip, and then the combustor tip support on the sensor housing. (Figure 17)

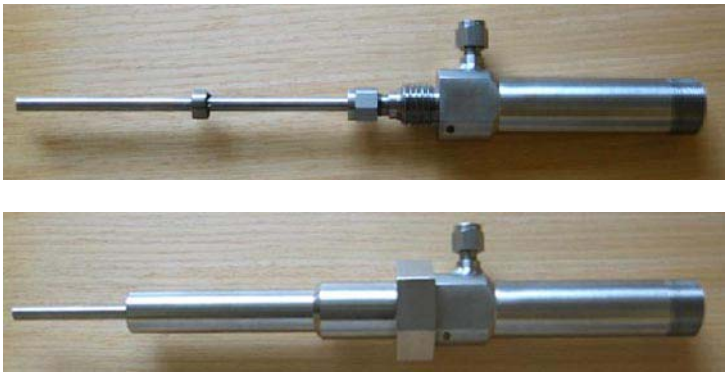


Figure 17

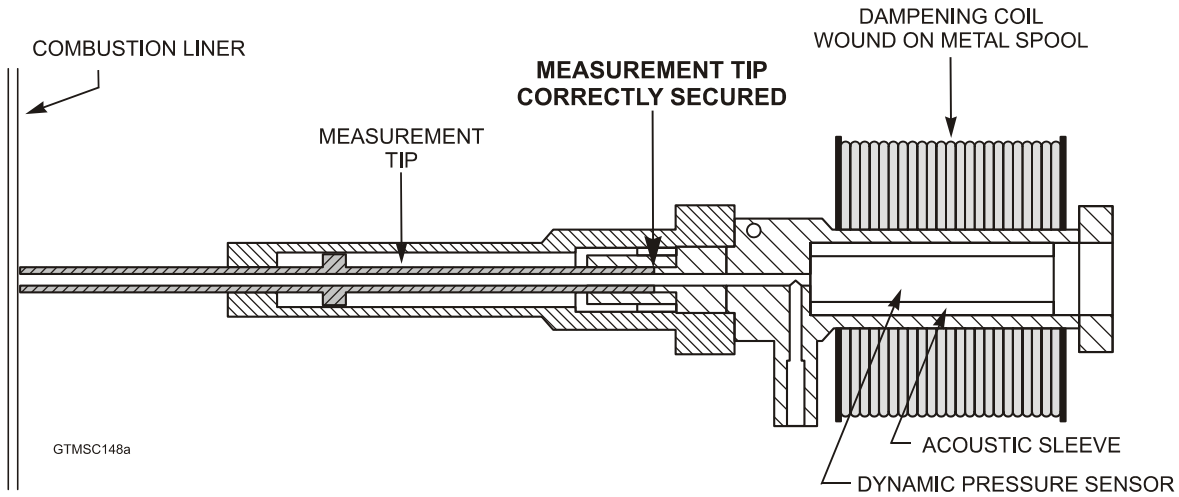
Take the partially assembled Gen II Probe holder shown in Figure 18 and the dynamics probe setting tool identified in the reference documents section as reference 1. Review the instructions located within Reference Document 2 (Procedure for Using the Dynamics Probes Depth Gauge) prior to installing the Dampening Coil. This will allow for easier handling of the assembly since the Dampening Coil is relatively heavy.



Figure 18

Make sure that the measurement tip is correctly installed and the combustor tip support is tight before using the Dynamics Probe Setting Tool (Reference Document 1) to tighten the ¾" Swagelok Fitting. See Figure 19.

Measurement Tip Correctly Secured



Measurement Tip Incorrectly Secured

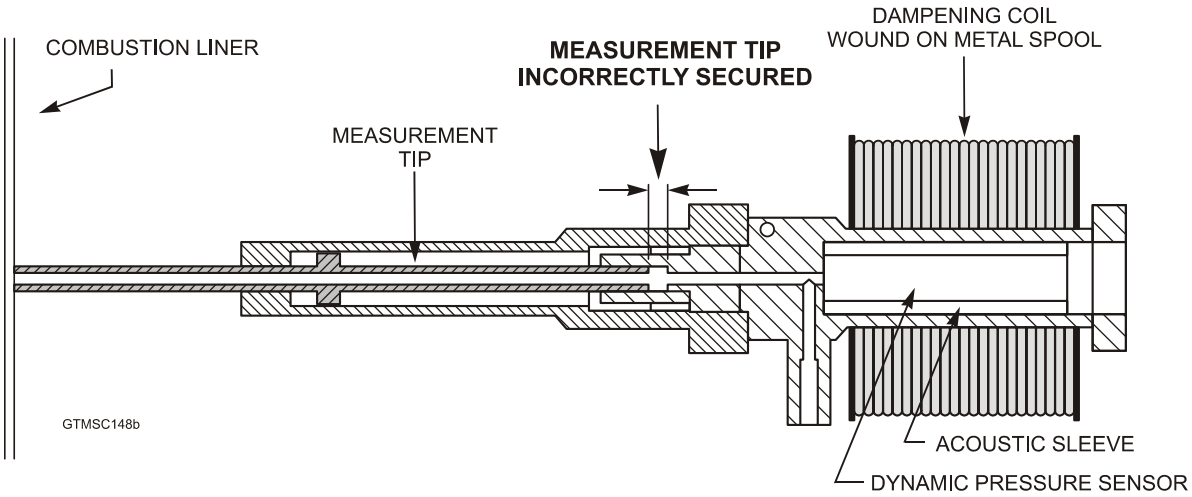


Figure 19

REVISION PAGE

Original Release Date
October 14, 2004

Revision Date	Revision	Revision Details	Approved By
10/14/04	–	New EM Article	MJC
10/21/04	A	Torque value on page 4, Step 7 changed from 80 ft-lbs to 70 ft-lbs	MJC
01/26/06	B	Caution statement added to page 6. Updated Unit Record Sheet Site URL	GTM